

Lab. 12 - Project Report

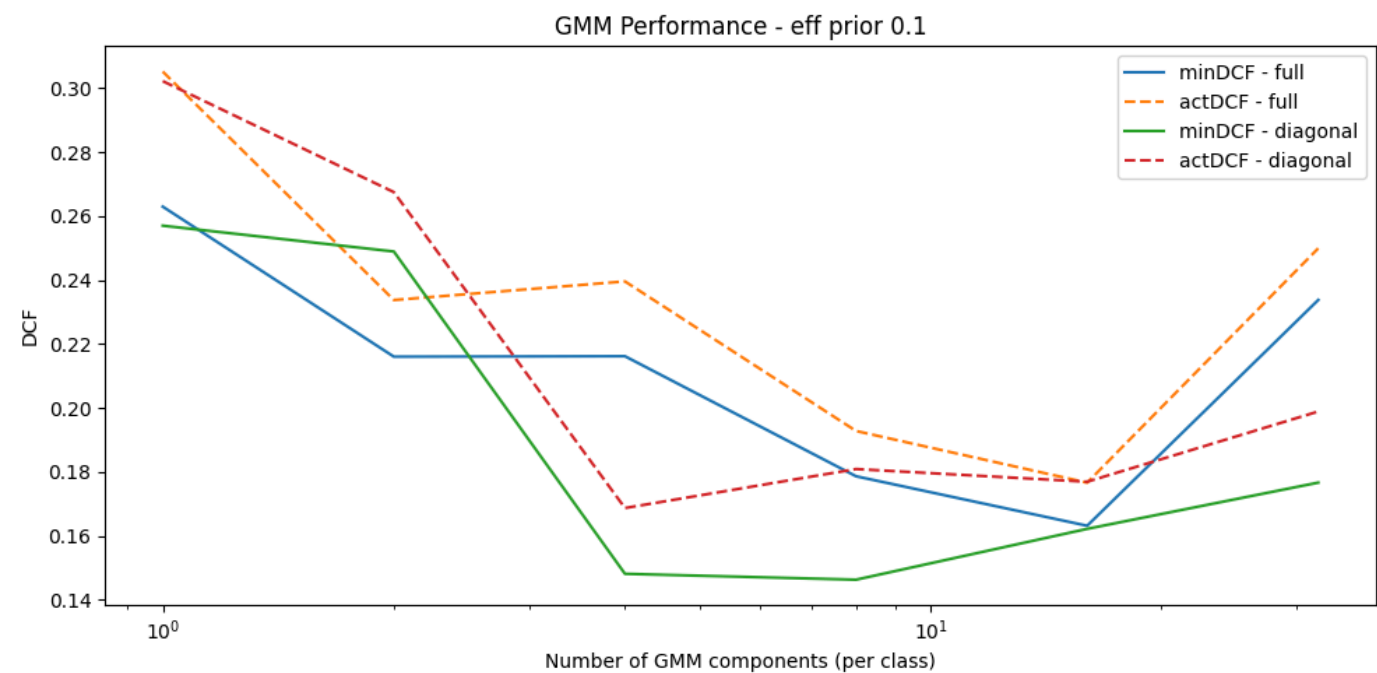
GMM Classification Report

Model Selection and Evaluation

We trained Gaussian Mixture Models (GMMs) on the binary classification task using the following configurations:

- **Covariance types:** full , diagonal
- **Number of components per class:** 1, 2, 4, 8, 16, 32
- **Application:** Target prior $\pi_T = 0.1$
- **Metrics:** Minimum DCF and actual DCF

The results are visualized below:



Summary Table

# Components	minDCF (full)	actDCF (full)	minDCF (diag)	actDCF (diag)
1	0.263	0.305	0.257	0.302
2	0.216	0.234	0.249	0.267
4	0.216	0.240	0.148	0.169
8	0.179	0.193	0.146	0.181
16	0.163	0.177	0.162	0.177
32	0.234	0.250	0.177	0.199

Observations

Full vs Diagonal Covariance

- **Diagonal models outperform full models** in terms of both minimum and actual DCF for most configurations.
- Diagonal models with 4–8 components reach the best overall performance (e.g., minDCF = 0.146 with 8 components).

Effect of Component Count

- Increasing the number of components generally improves performance up to 8–16 components.
- Beyond 16 components, performance begins to degrade slightly, likely due to overfitting, especially for full covariance models.

Calibration

- Actual DCFs tend to follow the trend of minDCF closely, indicating reasonably good calibration of the models.
- However, full covariance models show slightly higher actual DCFs than their minimum DCF, suggesting some calibration loss at higher complexity.

Conclusions

- **Best model:** Diagonal GMM with 8 components per class.
- **Key advantages:**
 - Simpler covariance structure helps prevent overfitting.
 - Lower computational cost compared to full models.
 - More robust and better-calibrated in practice.